

Use Cases AWS Cloudtrail

1. A log metric filter and alarm should exist for usage of the "root" user.

Path: /var/ossec/etc/rules/local_rules.xml

```
<group name="root_user">
  <rule id="100001" level="5">
    <decoded_as>json</decoded_as>
    <field name="user">root</field>
  </rule>
</group>
```

For alarm:

```
<group name="root_user">
  <rule id="100001" level="5">
    <decoded_as>json</decoded_as>
    <field name="user">root</field>
    <description>Root User Activity Detected</description>
  </rule>
</group>
```

2. Ensure log metric filter and alarm exist for security group changes.

```
<group name="security_group_changes">
  <rule id="100002" level="5">
    <decoded_as>json</decoded_as>
    <field name="event.action">modify_security_group</field>
  </rule>
```

</group>

```
<group name="security_group_changes">
  <rule id="100002" level="5">
    <decoded_as>json</decoded_as>
    <field name="event.action">modify_security_group</field>
    <description>Security Group Change Detected</description>
  </rule>
</group>
```

3. 'Ensure log metric filter and alarm exist for changes to network access control lists (NACL).'

```
<group name="nacl_changes">
  <rule id="100003" level="5">
    <decoded_as>json</decoded_as>
    <field name="event.action">modify_nacl</field>
  </rule>
</group>
```

```
<group name="nacl_changes">
  <rule id="100003" level="5">
    <decoded_as>json</decoded_as>
    <field name="event.action">modify_nacl</field>
    <description>NACL Change Detected</description>
  </rule>
```

```
</group>
```

4. Ensure a log metric filter and alarm exist for changes to network gateways

To monitor changes to network gateways in AWS CloudTrail logs, we have to create a custom rule for this:

```
<group name="aws,">
  <rule id="100001" level="7">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>network gateway</matches>
    </field>
    <description>Network Gateway Change Detected</description>
  </rule>
</group>
```

5. Ensure a log metric filter and alarm exist for route table changes:

We have to add a custom rule definition to monitor CloudTrail logs for route table changes:

```
<group name="aws">
  <rule id="100003" level="7">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>ec2.amazonaws.com CreateRoute | DeleteRoute | ReplaceRoute |
CreateRouteTable | DeleteRouteTable</matches>
    </field>
```

```
<description>Route Table Change Detected</description>
</rule>
</group>
```

6. Ensure a log metric filter and alarm exist for VPC changes.

We have to set up AWS CloudTrail to capture these events and then configure Wazuh to monitor and alert on the CloudTrail logs:

```
<group name="aws">
  <rule id="100004" level="7">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>ec2.amazonaws.com CreateVpc | DeleteVpc | ModifyVpcAttribute</matches>
    </field>
    <description>VPC Change Detected</description>
  </rule>
</group>
```

7. Ensure a log metric filter and alarm exist for unauthorized API calls.

We have to set up CloudTrail to capture API events, creating a CloudWatch Logs Metric Filter, and configuring an alarm. Additionally, you can create a corresponding rule in Wazuh to provide an additional layer of monitoring and alerting:

```
<group name="aws">
  <rule id="100005" level="10">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>unauthorized access|insufficient permissions|access denied</matches>
    </field>
    <description>Unauthorized API Call Detected</description>
  </rule>
</group>
```

8. Ensure a log metric filter and alarm exist for management console sign-in without MFA.

We have to set up CloudTrail to capture sign-in events, creating a CloudWatch Logs Metric Filter, and configuring an alarm. Additionally, you can create a corresponding rule in Wazuh for an extra layer of monitoring and alerting:

```
<group name="aws">
  <rule id="100006" level="10">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>signin.amazonaws.com ConsoleLogin (Failure|Success) signin:failure</matches>
    </field>
    <description>Management Console Sign-In Without MFA Detected</description>
  </rule>
</group>
```

9. Ensure a log metric filter and alarm exist for IAM policy changes.

We have to set up CloudTrail to capture policy events, creating a CloudWatch Logs Metric Filter, and configuring an alarm. Additionally, you can create a corresponding rule in Wazuh for an extra layer of monitoring and alerting:

```
<group name="aws">
  <rule id="100007" level="10">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>iam.amazonaws.com PutGroupPolicy | PutRolePolicy | PutUserPolicy | DeleteGroupPolicy
| DeleteRolePolicy | DeleteUserPolicy | CreatePolicyVersion | DeletePolicyVersion</matches>
    </field>
    <description>IAM Policy Change Detected</description>
  </rule>
</group>
```

```
</rule>  
</group>
```

10. Ensure a log metric filter and alarm exist for cloudtrail configuration.

We have to set up CloudTrail to capture configuration change events, creating a CloudWatch Logs Metric Filter, and configuring an alarm. Additionally, you can create a corresponding rule in Wazuh for an extra layer of monitoring and alerting:

```
<group name="aws">  
  <rule id="100008" level="10">  
    <decoded_as>json</decoded_as>  
    <field name="log">  
      <matches>cloudtrail.amazonaws.com UpdateTrail | CreateTrail | DeleteTrail | StartLogging |  
StopLogging | PutEventSelectors | DeleteEventSelectors</matches>  
    </field>  
    <description>CloudTrail Configuration Change Detected</description>  
  </rule>  
</group>
```

11. Ensure a log metric filter and alarm exist for AWS management console authentication failures.

We have to set up CloudTrail to capture authentication failure events, creating a CloudWatch Logs Metric Filter, and configuring an alarm. Additionally, you can create a corresponding rule in Wazuh for an extra layer of monitoring and alerting:

```

<group name="aws">
  <rule id="100009" level="10">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>signin.amazonaws.com Console Login (Failure) signin:failure</matches>
    </field>
    <description>Management Console Authentication Failure Detected</description>
  </rule>
</group>

```

12. Ensure a log metric filter and alarm exist for disabling or scheduled deletion of customer created CMKs.

AWS Key Management Service (KMS), We have to set up CloudTrail to capture relevant events, creating a CloudWatch Logs Metric Filter, and configuring an alarm. Additionally, you can create a corresponding rule in Wazuh for an extra layer of monitoring and alerting:

```

<group name="aws">
  <rule id="100010" level="10">
    <decoded_as>json</decoded_as>
    <field name="log">
      <matches>kms.amazonaws.com DisableKey | ScheduleKeyDeletion</matches>
    </field>
    <description>CMK Disabling or Scheduled Deletion Detected</description>
  </rule>

```

</group>